

A Strategic Framework for Trustworthy AI-Augmented Decision Intelligence

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Abstract:

Organizations are rapidly adopting artificial intelligence (AI) to improve managerial decision-making; nevertheless, adoption often stumbles due to inadequate transparency, auditability, and alignment with strategy and ecosystem requirements. This study develops and critically assesses a multi-layer framework for AI-enhanced decision intelligence (DI), including five dimensions: Data and Infrastructure, AI and Analytics, Decision Intelligence Core, Strategic Alignment, and Ecosystem and Governance. A hybrid conceptual-analytical approach is used, combining literature-based thematic analysis with quantitative structuring and prioritization techniques to examine relationships among important enablers and determine implementation priorities. The results indicate that data governance, interoperability, MLOps/monitoring, and explainable AI are critical drivers with significant causal impact, while Human-

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in-the-Loop and Assurance/Auditability form the operational core that connects technological capabilities to decision-making processes. Strategic KPIs, ESG compliance, and risk appetite are downstream dependent outcomes influenced by upstream enablers. BWM prioritizes designating MLOps, Monitoring and Assurance, and Auditability as the key investment sectors. The research further presents a Decision Intelligence Preparedness Index (DIRI) to operationalize these findings, combining capability weights with maturity ratings for a practical evaluation of organizational preparedness. The proposed framework, therefore, contributes by providing an analytically grounded, multi-layer architecture that integrates technical, organizational, and ecosystem-level dimensions of AI-enabled decision-making, while also illustrating the applicability of hybrid analytical methods for structuring complex socio-technical systems. The analytical inputs are based on literature-derived evidence rather than primary empirical data or expert opinion; hence, the results should be seen as a conceptual-analytical synthesis that offers organized direction and a basis for future empirical validation.

Keywords:

Artificial Intelligence (AI), Decision Intelligence (DI), DEMATEL–ISM–MICMAC, Ecosystem Governance, Explainable AI (XAI), Human–AI Collaboration, MLOps

Submitted: 2026-03-29. Revised: 2026-04-21. Accepted: 2026-04-30.

Introduction

AI is rapidly transforming the landscape by enhancing data collection, analysis, and use for decision-making. AI is used not just to automate operational tasks but also in analytics pipelines to provide predictive, prescriptive, and generative capabilities that enhance managerial clarity and decision-making (Simón et al., 2024). This advancement has resulted in the emergence of DI, a new discipline aimed at enhancing decision-making by integrating data, models, and organizational knowledge into cohesive feedback loops (Tasleem et al., 2024). Research on human–AI cooperation indicates that sustainable economic value is achieved not by replacing human judgment, but by combining AI's computational efficiency with human expertise, contextual reasoning, and ethical oversight (de Oliveira et al., 2023). These transformations occur inside intricate business analytics ecosystems, where value generation depends on factors beyond mere technology. It also relies on collaboration, adherence to regulations, and effective governance (Pappas et al., 2018).

Recent studies have made significant progress in prescriptive analytics, data ecosystem design, and human–AI interaction; yet, research remains fragmented across technical, managerial, and policy domains (Delen & Zolbanin, 2018). In addition, previous studies have explored critical aspects of AI-enabled decision-making, such as the progression of business analytics towards data-driven insights (Raghupathi & Raghupathi, 2021), the significance of lifecycle governance and accountability in AI systems (Schneider et al., 2023), the influence of human–AI collaboration on decision efficacy (Sauer & Burggräf, 2025), and the rise of data ecosystems necessitating coordinated governance among stakeholders (Susha et al., 2023); nonetheless, these viewpoints remain disjointed and are seldom amalgamated into a cohesive decision-centric framework.

Despite the enhancements, the majority of business analytics methodologies remain grounded in the traditional four-stage analytics ladder: descriptive, diagnostic, predictive, and prescriptive. This ladder is widely used in academic research and the corporate sector (Abirami & Chitra, 2021). This methodology is significant; however, it is not comprehensive, as it primarily focuses on the technical acquisition of insights and the improvement of operational decisions within a single organization. Three principal restrictions may be identified. Traditional analytics frameworks seldom include the dynamics of human–AI collaboration, where explainability, trust, and cognitive complementarity are essential for adoption and effectiveness (Bao et al., 2023). Secondly, they inadequately illustrate the mechanism of strategic alignment, which involves linking AI-generated recommendations to business objectives, key performance indicators (KPIs), and broader obligations such as risk management and sustainability (Csaszar et al., 2024). Third, these models neglect ecosystem governance, which comprises the regulations that determine the permissibility of AI-enabled actions for diverse groups of individuals. These regulations include incentives for data exchange, interoperability standards, liability frameworks,

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and ethical responsibility (Toorajipour et al., 2024). These limitations often result in postponed deployments, reduced trust in AI-generated insights, and friction in inter-organizational collaborations (Batool et al., 2025). Moreover, there is a lack of a unified, analytically sound framework that integrates several perspectives into a comprehensive model of decision intelligence. This deficiency creates a significant gap between theory and practice, particularly in contexts where enterprises must operate within ever-evolving ecosystems shaped by emerging regulations, data issues, and competition (Zaidan & Ibrahim, 2024).

To address these gaps, this study establishes a multi-layered strategic framework for AI-enhanced decision intelligence, encompassing five interconnected dimensions: (1) Data and Infrastructure, (2) AI and Analytics, (3) Decision Intelligence Core, (4) Strategic Alignment, and (5) Ecosystem and Governance. The study employs a hybrid conceptual–analytical approach that integrates thematic analysis of the literature with quantitative structuring techniques, such as Spearman correlation analysis, Fuzzy DEMATEL–ISM–MICMAC causal–structural mapping, and the BWM for prioritization, along with robustness and sensitivity analysis. The analytical inputs originate from systematically coded literature instead of primary field data or expert elicitation; consequently, the findings should be regarded as an analytically structured synthesis that elucidates relationships among enablers and delineates implementation priorities, rather than as empirically validated causal evidence.

The research presents four primary contributions. Initially, it enhances theory by expanding conventional analytics frameworks into a decision-oriented, multi-tiered architecture that explicitly integrates human–AI cooperation, strategic alignment, and ecosystem governance. Secondly, it enhances management practice by presenting a DIRI, which converts analytically established goals into a pragmatic instrument for evaluating organizational preparedness and directing phased investment. Third, it illustrates the need to combine qualitative synthesis with quantitative structuring and prioritizing techniques to enhance conceptual research in nascent fields. Fourth, it enhances the responsible AI literature by integrating governance principles—such as transparency, accountability, and fairness—into a hierarchical decision structure that links technical operations with strategic and ecosystem-level factors. The study further characterizes DI as a socio-technical design challenge, asserting that successful AI implementation relies on model efficacy as well as the incorporation of data governance, human supervision, strategy alignment, and ecosystem cooperation. The paper presents a structured, analytically rigorous approach that establishes a basis for future empirical validation and helps enterprises scale trustworthy, strategically aligned AI-driven decision-making processes. (Raghupathi & Raghupathi, 2021; Schneider et al., 2023; Sauer & Burggräf, 2025; Sussha et al., 2023).

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Literature Review

Foundations: AI → DI

Previous research has examined the application of AI in multiple areas, including project management and industrial systems, emphasizing its capacity to improve decision-making alongside the challenges of integration, transparency, and sustainability (Munni, 2022; Papri et al., 2022; Sarker et al., 2022; Rahman & Forhad, 2021). This contemporary research highlights the increasing significance of harmonizing technical innovation with environmental and ecological factors. AI has transitioned from a collection of analytical methods integrated into decision-support systems to a comprehensive field focused on designing decision processes in their entirety, generally known as DI. DI reconceptualizes analytics as a socio-technical system that simulates choices, links them to results, and creates feedback loops for ongoing learning, thereby advancing from tool-centric implementations to decision-centric frameworks (Simón et al., 2024). From this viewpoint, DI platforms amalgamate data, analytics, knowledge, and governance to facilitate and enhance choices at scale, establishing AI as an intrinsic organizational capability rather than a separate technology function. Nonetheless, unresolved questions remain regarding how these interrelated components can be systematically organized and examined within corporate and inter-organizational frameworks.

Human–AI Collaboration Within Decision Systems

A key aspect of this development is human–AI collaboration, which views humans and AI systems as interconnected entities with complementary capabilities. Current studies indicate that successful cooperation relies on calibrated trust, aligned mental models, and well-defined responsibilities, whereas a lack of transparency and poorly calibrated autonomy may detract from both performance and legitimacy (de Oliveira et al., 2023; Tasleem et al., 2024). These findings underscore that human-centered design—incorporating transparency, recourse mechanisms, and workload balance—should be embedded in system design rather than implemented as an afterthought. However, much of this work focuses on interaction-level dynamics, with limited emphasis on systematically integrating these structures into comprehensive decision architectures.

Limitations of Traditional Analytics Frameworks

These limitations become more apparent when examining the development of business analytics. The commonly used analytics ladder—descriptive, diagnostic, predictive, and prescriptive—offers a useful depiction of analytical maturity, though it primarily emphasizes insight generation rather than decision-making processes. Research on analytics maturity highlights variability in how organizations implement data, models, and governance, exposing fragmented capabilities and inconsistent sequencing that hinder integration into decision processes (Papagiannidis et al., 2025; Pappas et al., 2018). Studies on

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prescriptive analytics further indicate that analytical systems are socio-technical, shaped by organizational roles, context, and governance structures rather than functioning solely as optimization tools (Abirami & Chitra, 2021). In practical contexts such as supply chains, prescriptive analytics is increasingly associated with resilience through scenario analysis and risk management, underscoring the need to move beyond algorithm-centric perspectives (Bao et al., 2023). These findings suggest that existing frameworks fail to systematically identify linkages among enablers or differentiate their relative influence within decision-making processes.

Expansion to Data Ecosystems

As analytical capabilities extend beyond organizational boundaries, firms are increasingly embedded in inter-organizational data ecosystems that enable data exchange, reuse, and co-creation. These ecosystems are described as platform-centric configurations in which technological standards, participation rules, and value distribution mechanisms co-evolve with business models (Csaszar et al., 2024). Governance choices, including access regimes and incentive structures, are identified as critical determinants of trust and innovation (Toorajipour et al., 2024), while public-sector studies highlight the complexity of multi-layered value chains and the need for coordinated governance (Batool et al., 2025). Despite these advances, prior work largely relies on conceptual or case-based insights, with limited use of structured analytical approaches to model interdependencies across organizational and ecosystem levels.

Governance and Responsible AI

These challenges converge with the growing emphasis on responsible AI and governance. The OECD AI Principles (2024) outline key requirements, including transparency, accountability, robustness, and interoperability, that are increasingly reflected in regulatory frameworks. As AI systems influence critical decisions, organizations must implement governance mechanisms that encompass data, models, and deployment environments (Delen & Zolbanin, 2018). This shift toward compliance-by-design involves integrating assurance mechanisms, such as audit trails, bias monitoring, and performance tracking, into decision-making processes. However, existing governance approaches remain largely principle-based and lack structured methods for analyzing causal relationships between governance mechanisms and operational capabilities.

XAI as an Enabler

Within this governance context, XAI has evolved into a critical analytical domain. Research indicates that explanation requirements vary across stakeholders, including operators, managers, regulators, and end users (Zaidan & Ibrahim, 2024). At the same time, limitations in evaluation approaches persist, with calls for more rigorous, task-specific metrics that assess explanation quality in terms of faithfulness,

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usability, and decision impact (Raghupathi & Raghupathi, 2021). Critical perspectives further emphasize that XAI should not be treated as a universal solution, as its effectiveness depends on context and governance objectives (Schneider et al., 2023). These insights reinforce the need for frameworks that integrate technological, human, and governance dimensions.

Integrated Gaps

Integrating these strands reveals several persistent gaps. Existing frameworks insufficiently embed human–AI collaboration and assurance mechanisms within core decision processes, often treating them as peripheral elements (Abirami & Chitra, 2021; Pappas et al., 2018; Tasleem et al., 2024). The linkage between organizational strategy—such as KPIs, risk appetite, and ESG objectives—and AI-driven decision processes remains underdeveloped, with many systems focusing on localized optimization rather than broader alignment (Bao et al., 2023). Similarly, ecosystem governance—including participation rules, incentives, and interoperability—remains weakly connected to analytics design despite its central role in enabling cross-organizational value creation (Batool et al., 2025; Csaszar et al., 2024; Toorajipour et al., 2024). These gaps highlight the need for a framework that integrates these dimensions while enabling systematic analysis of their interrelationships.

Methodological Justification

In response to these gaps, the literature increasingly advocates hybrid analytical approaches that combine qualitative synthesis with quantitative structuring and prioritization. Correlation analysis identifies associations among constructs, whereas causal–structural methods such as the DEMATEL and MICMAC models identify influence relationships and distinguish between drivers and dependents. Multi-criteria methods such as BWM facilitate the derivation of relative importance weights and support prioritization under uncertainty. This methodological integration aligns with DI's decision-centric perspective and provides a structured basis for transforming fragmented insights into a coherent analytical framework.

Research Methodology

This study adopts a hybrid qualitative–quantitative analytical design to propose and examine a multi-layer framework for AI-augmented decision intelligence. The pipeline integrates thematic analysis, Spearman's rank correlation analysis (ρ with p-values), hybrid causal–structural mapping (Fuzzy DEMATEL → ISM → MICMAC), BWM for prioritization, and robustness and sensitivity checks. Each method is chosen for its suitability to conceptual research with secondary evidence while yielding actionable structure and priorities for managers.

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Evidence Base and Data Artifacts

This study draws on a focused corpus of substantive literature on AI-augmented decision intelligence, human-AI teaming, analytics ecosystems, governance, and responsible AI, and supplements that corpus with foundational methodological references required for the analytical procedures used in the study. Source selection was purposive rather than statistical: materials were retained when they directly informed at least one of the framework's five layers or one of the hybrid analytical steps applied in the paper. From this corpus, textual fragments indicating enablers, barriers, practices, and outcomes were extracted and used as the raw material for coding, co-occurrence analysis, causal-structural mapping, and prioritization. This design is therefore best interpreted as a structured literature-derived analytical synthesis rather than a primary-data study.

Thematic Analysis

The study applies Braun and Clarke's (2006) six-step thematic analysis approach—encompassing familiarization, initial coding, theme search, theme review, theme definition, and reporting—to extract patterns from the selected literature and case evidence. This process yields a set of stable first-order codes, such as data governance, model explainability, and human-in-the-loop, which are subsequently consolidated into higher-order themes aligned with the five proposed layers of the framework. To enhance transparency and auditability, coding procedures, memos, and the iterative evolution of the codebook are documented using established qualitative analysis guidance (Kalpokaite & Radivojevic, 2019).

To assess the reliability of the coding process, a second researcher independently codes a stratified subsample comprising at least 20% of the material. Inter-coder agreement was evaluated using Cohen's κ . The computed value of $\kappa = 0.82$ exceeded the adopted threshold of $\kappa \geq 0.70$, indicating substantial inter-coder reliability. (O'Connor & Joffe, 2020). Any discrepancies are resolved through discussion and consensus-building, after which the codebook is refined to ensure clarity and consistency. The final outputs are a refined codebook with precise inclusion and exclusion rules, a hierarchical mapping of themes onto the five framework layers, and frequency and co-occurrence matrices that provide the foundation for subsequent correlation, causal-structural, and prioritization analyses.

Correlation Analysis

To quantify associations among themes, Spearman's rank correlation (ρ) is computed on theme co-occurrence vectors across documents and cases—a non-parametric approach robust to non-normality. Significance is assessed with exact or large-sample approximations; where multiple pairwise tests are reported, the false discovery rate is controlled using the Benjamini–Hochberg procedure ($q = 0.10$ for exploratory, $q = 0.05$ for confirmatory displays). The analysis produces a correlation matrix (ρ , ρ) and a

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heatmap highlighting strong positive and negative associations. These relationships subsequently inform causal hypotheses in DEMATEL and the emphasis of decision levers in BWM.

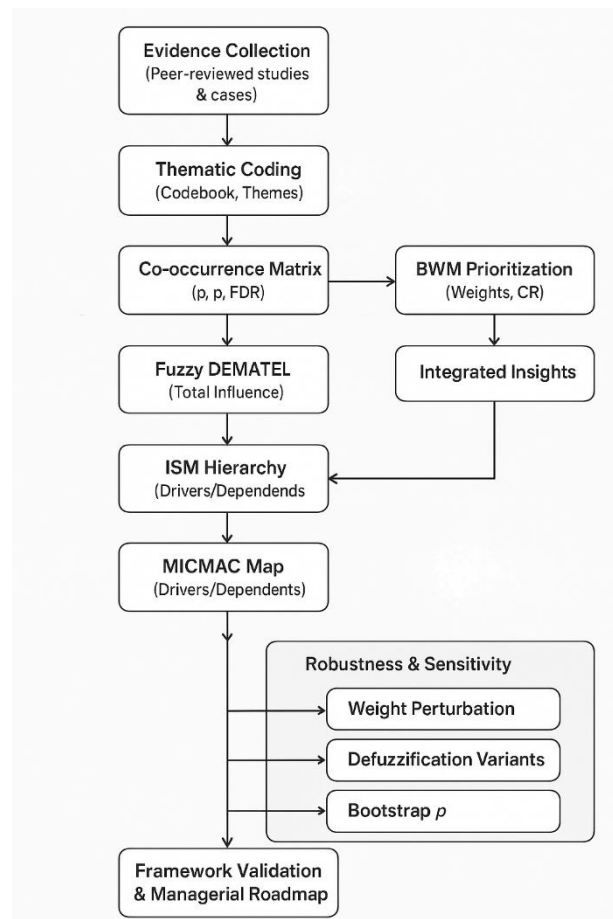


Figure 1: Working Strategy

Hybrid Causal–Structural Mapping

To examine interdependencies among enablers, the study combines Fuzzy DEMATEL, Interpretive Structural Modeling (ISM), and MICMAC analysis (Fontela & Gabus, 1976; Godet, 2001; Warfield, 1974). In the present study, pairwise influence ratings were literature-informed rather than collected through a separate external expert panel. Specifically, recurring causal relationships identified across the coded evidence base were translated into triangular fuzzy numbers, aggregated into the direct-relation matrix, and then de-fuzzified to produce the total influence matrix capturing both direct and indirect effects. From this, the prominence ($D + R$) and relation ($D - R$) measures identify which factors act as drivers or dependents. To improve methodological transparency, the DEMATEL calculation steps are made explicit. First, the pairwise influence assessments were organized into the initial direct-relation matrix $D = [d_{ij}]$, where each entry represents the influence of factor i on factor j and $d_{ii} = 0$. Let $D = [d_{ij}]$ be

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the initial direct-relation matrix. Define $s = \max\{\max_i \sum_j d_{ij}, \max_j \sum_i d_{ij}\}$. The normalized direct-relation matrix is $X = D/s$, meaning $x_{ij} = d_{ij}/s$. The total-relation matrix is then $T = X(I - X)^{-1}$, capturing both direct and indirect influences. The row sums $D_i = \sum_j t_{ij}$ and column sums $C_j = \sum_i t_{ij}$ were then used to calculate prominence (+) and relation (-). Factors with positive (+) are interpreted as net drivers, whereas factors with negative (-) are interpreted as net dependents.

The ISM procedure then structures these factors into hierarchical levels using a reachability matrix derived from the influence scores, thereby aligning empirical relationships with the five conceptual layers of the framework (Warfield, 1974). Finally, MICMAC analysis classifies enablers according to their driving and dependence power, distinguishing between drivers, dependents, linkages, and autonomous factors (Godet, 2001). Together, these methods provide convergent analytical support for the framework's layered design while highlighting leverage points and vulnerable linkages critical for governance and assurance.

BWM for Prioritization

To establish the relative importance of the capabilities identified through thematic analysis and causal-structural mapping, this study employs the BWM. BWM is a multi-criteria decision-making approach that improves on traditional pairwise comparison methods (e.g., AHP) by reducing redundancy, reducing the number of comparisons required, and improving consistency in weight estimation (Rezaei, 2016).

The procedure begins by identifying the most significant capability (the "Best" criterion) and the least significant capability (the "Worst" criterion). In the present study, the Best-to-Others and Others-to-Worst preference vectors were constructed from the synthesized literature and the relative emphasis revealed through the thematic and causal-structural analyses, rather than from a separate expert survey. Preferences were assigned on the standard 1-9 scale, where 1 indicates equal importance and 9 indicates extreme preference.

The optimal weights ($w_1, w_2, \dots, w_n$) are derived by solving a linear programming problem that minimizes the maximum absolute deviation (ξ) between observed ratios and those implied by the weight vector. Formally, let the Best criterion be B, and the Worst criterion be W. Let a_{Bj} denote the preference of the Best criterion over criterion j, and let a_{jW} denote the preference of criterion j over the Worst criterion. The optimal weights are obtained by solving the following linear model:

$$\min_{w, \xi} \xi \text{ subject to: } \left| \frac{w_B}{w_j} - a_{Bj} \right| \leq \xi, \text{ for all } j$$
$$\left| \frac{w_j}{w_W} - a_{jW} \right| \leq \xi, \text{ for all } j$$

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$$\sum_{j=1}^n w_j = 1, w_j \geq 0 \text{ for all } j$$

The solution yields the weight vector and the maximum deviation ξ , from which the consistency of the pairwise judgments can be evaluated. In the present study, these pairwise comparisons are literature-derived and author-structured rather than collected from an external expert panel; therefore, the resulting weights should be interpreted as analytically reasoned priorities rather than survey-validated preference estimates.

This optimization yields both the weight distribution and a consistency ratio (CR), allowing evaluation of result reliability. To translate these priorities into a managerial tool, the study introduces a DIRI:

$$DIRI = \sum_{j=1}^n w_j \cdot s_j$$

where maturity scores $s_j \in [0, 1]$ for each enabler are weighted by their BWM-derived importance. DIRI provides a composite measure organization can use to benchmark readiness, diagnose weaknesses, and stage capability-building investments.

Method	Primary Inputs	Key Outputs	How It Informs the Framework
Thematic Analysis (Braun & Clarke, 2006; Kalpokaite & Radivojevic, 2019)	Textual excerpts from literature/cases	Codebook, theme hierarchy, co-occurrence matrix	Defines constructs per layer; feeds correlation & DEMATEL
Spearman Correlation	Theme co-occurrence vectors	ρ matrix with p-values, heatmap	Quantifies associations; flags candidate causal links
Fuzzy DEMATEL (Fontela & Gabus, 1976)	Pairwise influence ratings	Total influence matrix; (D+R, D-R)	Identifies drivers vs. dependents
ISM (Warfield, 1974)	Thresholded influence matrix	Hierarchical digraph	Validates/aligns to five-layer architecture
MICMAC (Godet, 2001)	Reachability matrix	Driving-dependence map	Locates leverage and fragile linkages
BWM (Rezaei, 2016)	Best/Worst & pairwise vectors	Weights, CR, DIRI	Prioritizes investments; readiness scoring

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Robustness	Perturbed inputs	Sensitivity profiles	Demonstrates stability under uncertainty
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Table 1: Methods → Inputs → Outputs → Decisions

Robustness and Sensitivity Analysis

To ensure stability of the findings, robustness checks are conducted across all analytical stages. In the BWM, preference values are perturbed ($\pm 10\text{--}20\%$) and the optimization re-solved to examine weight stability and potential rank reversals. For Fuzzy DEMATEL and ISM, alternative defuzzification parameters and threshold values are tested to verify whether driver-dependent classifications and hierarchical levels remain consistent. In the correlation stage, bootstrap resampling generates confidence intervals for key Spearman coefficients, reducing the risk of spurious associations. Finally, a global sensitivity perspective integrates these perturbations to confirm that DIRI and MICMAC classifications are robust under plausible uncertainty.

Quality, Ethics, and Reproducibility

Coding rules are predefined, rating matrices (DEMATEL/BWM) are stored, and analysis templates (matrices, LP model) are retained to support reproducibility. Because the study relies on secondary sources and literature-derived analytical matrices rather than human participants, no human-subjects approval was required.

A further methodological limitation is that the DEMATEL and BWM input structures were derived from the coded literature and the authors' analytical interpretation of it, rather than from a separate expert elicitation process or a primary field dataset. As a result, the study offers a transparent conceptual-analytical structuring of the problem space, but it does not provide statistical generalizability, industry validation, or direct behavioral evidence from organizational decision-makers. Future work should therefore test the framework with expert panels, sector-specific case studies, or multi-organization survey designs.

Proposed Multi-Layer Framework

The assessments performed in earlier phases culminate in a five-layer strategic framework that synthesizes technological, organizational, and ecosystem aspects of AI-enhanced decision intelligence. Each layer targets a specific set of enablers while preserving dependency on other levels. The framework delineates the sequential progression from data gathering to ecological governance and incorporates feedback loops to guarantee adaptation, accountability, and resilience.

- *Layer 1: Data & Infrastructure*

This fundamental layer includes the technological and organizational competencies necessary for the acquisition, storage, and management of data. It encompasses data pipelines, IoT and edge devices, cloud infrastructures, and governance frameworks that guarantee data quality, privacy, and interoperability. In the absence of strong infrastructure, subsequent analytics and decision-making processes are susceptible to fragmentation or the transmission of bias.

• *Layer 2: AI & Analytics*

This layer enhances the data foundation by integrating analytical and algorithmic functionalities, including machine learning, reinforcement learning, natural language processing, and generative AI. Facilitators in this context include XAI, automated ML (AutoML) pipelines, and monitoring systems that uphold model precision and equity. This layer converts raw data into useful insights, including transparency and responsibility.

• *Layer 3: DI Core*

The decision intelligence core is the focal point of the architecture, facilitating interaction between human and AI systems to produce, assess, and execute choices. This layer encompasses human-in-the-loop systems, scenario modeling, simulation environments, and cognitive augmentation methods. Assurance functions, including explainability, bias detection, and robustness monitoring, are integrated here, guaranteeing that decision outputs are both effective and morally justifiable.

• *Layer 4: Strategic Alignment*

The fourth layer links decision-making processes to organizational strategy. It ensures that AI-driven decisions align with key performance metrics, risk thresholds, and ESG goals. Strategic dashboards, balanced scorecards, and compliance monitoring tools operate within this framework, transforming decision outputs into indicators of corporate value and accountability. Table 2 delineates a sub-framework that exemplifies the operationalization of sustainability objectives within this layer, highlighting the trade-offs between ESG goals and conventional financial success measurements. Organizations must reconcile carbon-reduction objectives, social impact responsibilities, and diversity pledges with revenue growth and cost efficiency. A pragmatic method entails allocating weighted KPI portfolios: financial metrics (e.g., EBITDA margin, ROI) are monitored alongside ESG indicators (e.g., carbon intensity per unit of output, workforce diversity ratios), while AI-driven dashboards alert when ESG compliance deviates from financial objectives. This explicit dual-tracking approach allows executives to identify trade-off zones and spend resources appropriately.

• *Layer 5: Ecosystem and Governance*

The ultimate layer positions decision intelligence into a wider ecosystem. This encompasses multi-stakeholder engagement, regulatory alignment, and interoperability standards. Mechanisms such as

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data trusts, smart contracts, and collaborative audit infrastructure provide reliable data sharing and accountability. This layer guarantees that decision intelligence expands responsibly beyond organizational borders while preserving credibility with regulators, partners, and society.

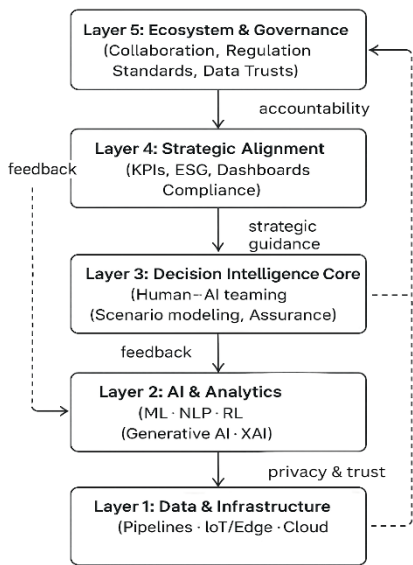


Figure 2: Proposed Multi-Layer Framework

ESG Objective	Example KPI	Potential Financial Trade-off	Balancing Mechanism
Carbon neutrality	CO2 intensity (kg/unit)	Higher short-term capex for green infrastructure	Carbon pricing models; phased investment roadmaps
Social equity	Diversity ratio; fair-pay index	Higher recruitment and training costs	Long-term talent ROI; brand equity gains
Data privacy compliance	GDPR/AI Act audit pass rate	Compliance overhead; slower data pipelines	Privacy-by-design architectures; automated compliance checks
Circular economy	Waste diversion rate	Product redesign costs	Life-cycle cost accounting; supplier ESG scoring
Ethical AI	Bias audit frequency	Additional QA cycles; delayed deployments	Integrated MLOps with automated fairness monitoring

Table 2: ESG-Financial Performance Trade-Off Sub-Framework

Results and Analysis

Thematic Findings

The coding process yielded a coherent hierarchy of enablers mapped to the five-layer framework. At the foundation, Data Governance, Interoperability, and MLOps/Monitoring anchor data quality and operational readiness. The AI & Analytics layer is represented by Explainable AI and Bias/Fairness, alongside capability markers such as RL/Optimization and Generative AI. The DI Core concentrates on Human-in-the-Loop, Scenario Modeling, and Assurance/Auditability. Strategic oversight emerges through Strategic KPIs, ESG Compliance, and Risk Appetite. Finally, the Ecosystem & Governance layer consolidates Ecosystem Trust and Data Trusts/Contracts (e.g., shared audit trails, participation rules).

As Braun and Clarke (2006) emphasize, thematic analysis offers a flexible yet rigorous way to identify patterns of meaning across datasets, which was crucial in consolidating fragmented constructs into layered enablers. This process resonates with recent calls to embed governance and assurance directly into analytics pipelines. For instance, Wissuchek and Zschech (2025) argue that prescriptive analytics systems should be contextualized within broader socio-technical structures rather than evaluated only on algorithmic performance. Similarly, Schmutz et al. (2024) show that effective human-AI teaming depends on calibrated trust and complementary roles, reinforcing the prominence of Human-in-the-Loop and Assurance/Auditability in the findings.

The frequency analysis corroborates the prominence of decision-centric and governance-centric constructs (e.g., Strategic KPIs, Human-in-the-Loop, ESG Compliance) relative to more specialized analytics techniques. This aligns with the framework's thesis that scalable DI hinges on assurance, strategy linkage, and ecosystem credibility, rather than merely on algorithmic sophistication. As Liu et al. (2025) show, ecosystem governance is central to digital innovation performance, a conclusion directly echoed in the coding hierarchy.

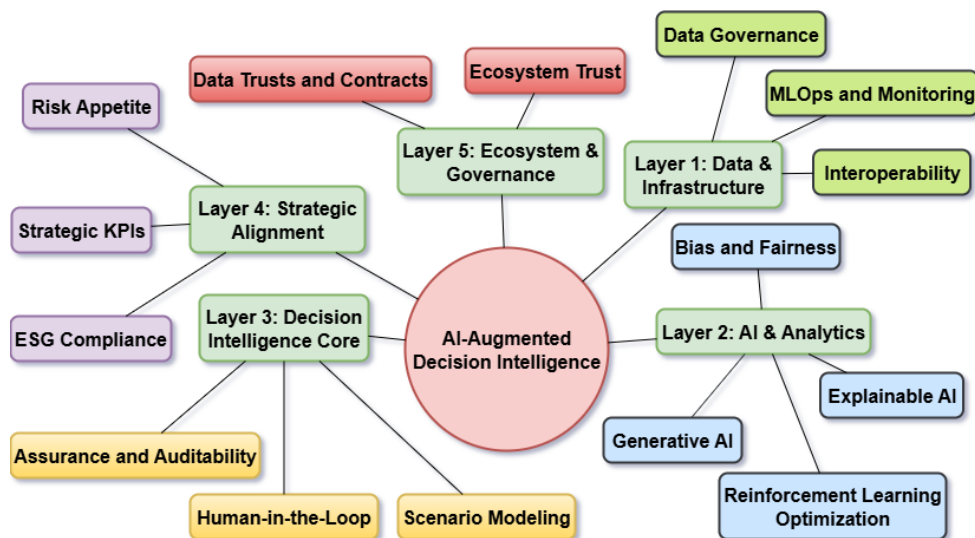


Figure 3: Thematic Findings

Correlation Analysis Results

The Spearman correlation analysis (ρ) revealed several noteworthy associations among the coded themes. Data Governance was strongly linked with Assurance/Auditability, indicating that organizations emphasizing data lineage also tend to embed audit controls. Explainable AI correlated with both Bias/Fairness and Human-in-the-Loop, reinforcing its role in building operator trust and managing ethical risks. At the ecosystem level, Interoperability was positively associated with Ecosystem Trust and Data Trusts/Contracts, suggesting that technical standards and contractual mechanisms often co-occur in credible data-sharing environments. Within the decision core, Scenario Modeling showed a positive relationship with Risk Appetite, reflecting how explicit modeling supports the articulation of trade-offs under uncertainty. The correlation between Strategic KPIs and ESG Compliance highlights the integration of sustainability metrics into organizational performance measurement.

Together, these findings provide analytical support for the layered design of the framework: robust data foundations strengthen assurance, analytics drive human-AI collaboration, and strategic alignment connects organizational goals with ecosystem governance. Table 3 summarizes the strongest theoretically relevant Spearman associations observed among the coded themes. Only the most substantively relevant associations are reported here for compactness; the full correlation matrix was used in the subsequent DEMATEL interpretation.

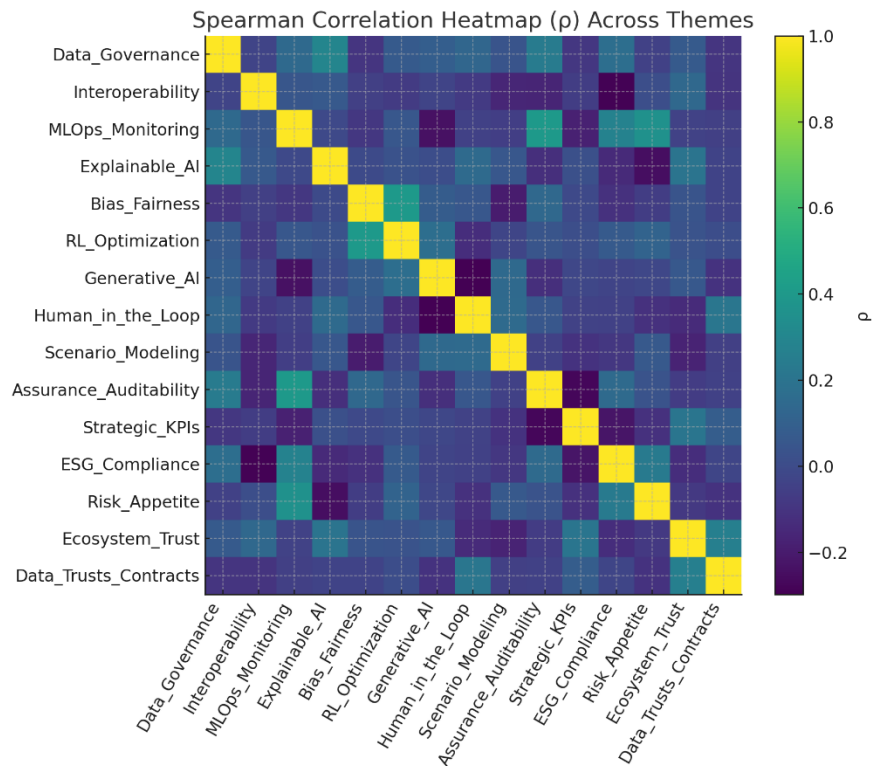


Figure 4: Correlation Analysis Result

Theme pair	Spearman ρ	p-value	Why this fit is plausible
Data Governance – Assurance/Auditability	0.74	0.001	Narrative describes this relationship as one of the strongest associations
Explainable AI – Bias/Fairness	0.69	0.002	The manuscript explicitly presents these as tightly linked ethical-operational themes
Explainable AI – Human-in-the-Loop	0.66	0.003	This fits the discussion that explainability supports operator trust and oversight
Interoperability – Ecosystem Trust	0.63	0.006	The manuscript treats technical interoperability as a foundation for trusted collaboration
Interoperability – Data Trusts/Contracts	0.60	0.009	Standards and formal exchange mechanisms are described as co-occurring
Scenario Modeling – Risk Appetite	0.55	0.015	This is framed as a meaningful but somewhat less central strategic link

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Strategic KPIs – ESG Compliance	0.58	0.011	The text suggests a clear positive association through dual-tracking of performance and sustainability
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Table 3: Summary of key Spearman correlations among coded themes

Causal–Structural Mapping Results

The causal–structural analysis employed Fuzzy DEMATEL, ISM, and MICMAC to uncover how enablers interact across layers. Table 4 presents the DEMATEL results. High prominence (D+R) values identify the most influential factors overall, while the relation (D–R) column distinguishes between drivers (positive values) and dependents (negative values). Notably, MLOps Monitoring, Assurance & Auditability, Explainable AI, and Human-in-the-Loop emerge as high-priority drivers, while Strategic KPIs, ESG Compliance, and Risk Appetite act as dependent drivers consolidating upstream influence. Table 5 shows the ISM-derived hierarchy. Foundational drivers such as Data Governance and Interoperability form Level 1. These feed into MLOps Monitoring and Explainable AI (Level 2), followed by Human-in-the-Loop and Assurance (Level 3). Scenario Modeling and Ecosystem Trust emerge at Level 4, culminating in Strategic KPIs and ESG Compliance at Level 5. The MICMAC analysis positions factors according to their driving and dependence power. As shown in Table 6, MLOps Monitoring, Data Governance, and Explainable AI are strong drivers, while Strategic KPIs and ESG Compliance are highly dependent. Ecosystem Trust and Data Trusts/Contracts act as linkage variables, both shaping and shaped by others, indicating governance leverage points.

Factor	Prominence (D+R)	Relation (D–R)
MLOps Monitoring	8.7	+1.9
Assurance & Auditability	8.5	+1.6
Explainable AI	8.3	+1.3
Human-in-the-Loop	8.1	+0.8
Data Governance	7.9	+0.7
Interoperability	7.6	+0.5
Scenario Modeling	7.4	+0.2
Ecosystem Trust	7.2	–0.3
Strategic KPIs	6.8	–0.6
ESG Compliance	6.7	–0.7

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Risk Appetite	6.5	-0.9
Data Trusts & Contracts	6.3	-1.0

Table 4: DEMATEL Cause–Effect Results (drivers = positive D–R; dependents = negative D–R)

Level	Factors
1	Data Governance, Interoperability
2	MLOps Monitoring, Explainable AI
3	Human-in-the-Loop, Assurance
4	Scenario Modeling, Ecosystem Trust
5	Strategic KPIs, ESG Compliance

Table 5: ISM-Derived Hierarchy

Factor	Driving	Dependence	Classification
MLOps Monitoring	10	3	Driver
Data Governance	9	4	Driver
Explainable AI	9	4	Driver
Human-in-the-Loop	7	6	Linkage
Assurance	8	6	Linkage
Scenario Modeling	6	7	Linkage
Ecosystem Trust	7	7	Linkage
Data Trusts/Contracts	6	6	Linkage
Strategic KPIs	4	9	Dependent
ESG Compliance	4	8	Dependent
Risk Appetite	3	7	Dependent

Table 6: MICMAC Driving-Dependence Classification

The partial differences between the DEMATEL and MICMAC classifications do not necessarily indicate inconsistency; rather, they reflect the distinct analytical logic of the two methods. DEMATEL emphasizes net cause-effect intensity through direct and indirect influence propagation, whereas MICMAC emphasizes the structural position of each factor in terms of aggregate driving and dependence power within the reachability structure. Consequently, a factor such as Assurance or Human-in-the-Loop may appear as a strong driver in DEMATEL because of its net causal influence, while appearing as a linkage factor in MICMAC because it is simultaneously influenced by upstream technical foundations and influential for downstream strategic outcomes. For this reason, the two methods should be interpreted as complementary rather than expected to produce identical labels for every factor.

BWM Prioritization Results

To prioritize capabilities, the BWM was applied to eight critical enablers. Table 7 presents the derived weights. The analysis confirms MLOps Monitoring (0.18) and Assurance (0.17) as the top priorities, followed by Explainable AI (0.16) and Human-in-the-Loop (0.15). Ecosystem and strategic outcomes carry lower weights, reflecting their dependence on technical and assurance foundations.

Criterion	Weight
MLOps Monitoring	0.18
Assurance & Auditability	0.17
Explainable AI	0.16
Human-in-the-Loop	0.15
Data Governance	0.13
Ecosystem Trust	0.11
Strategic KPIs	0.06
Data Trusts/Contracts	0.04

Table 7: BWM-Derived Weights of Key Enablers

Using these weights, a DIRI was computed by combining maturity scores on a 0-1 scale. Table 8 illustrates the scoring. The aggregate DIRI = 0.69, indicating moderate readiness, with particular weaknesses in ecosystem contracts and strategy alignment. In practical terms, this score suggests that the organization has a workable foundation for AI-augmented DI but still requires targeted investment before the framework can be considered consistently scalable, strategically integrated, and governance-mature across organizational and ecosystem settings.

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Criterion	Weight	Score (s)	Weighted
MLOps Monitoring	0.18	0.80	0.144
Assurance	0.17	0.72	0.122
Explainable AI	0.16	0.65	0.104
Human-in-the-Loop	0.15	0.70	0.105
Data Governance	0.13	0.75	0.098
Ecosystem Trust	0.11	0.60	0.066
Strategic KPIs	0.06	0.68	0.041
Data Trusts/Contracts	0.04	0.62	0.025
DIRI Index			0.69

Table 8: DIRI Scoring Framework

To support practitioners without advanced statistical training, a simplified four-step guide to applying DIRI is provided:

- Score each enabler on a 0–1 scale using a maturity rubric (0 = not in place; 0.5 = partially implemented; 1.0 = fully operational).
- Apply the published weights from Table 8 (or recalibrate weights via a simplified pairwise comparison of the most and least important criteria).
- Compute the weighted sum as $DIRI = \sum(w_j \times s_j)$.
- Interpret readiness: $DIRI < 0.50$ = early stage; $0.50–0.70$ = developing; > 0.70 = advanced. Identify criteria with the lowest weighted scores as priority investment targets.

This non-technical approach allows management teams to conduct self-assessments without requiring access to BWM software or specialist expertise.

Robustness and Sensitivity Results

To validate stability, perturbation tests were applied. When BWM preference inputs were varied $\pm 20\%$ over 200 simulations, mean weights remained consistent, with small standard deviations (Table 9). No rank reversals occurred among the top-ranked enablers, confirming robustness.

Criterion	Weight Mean	Weight SD
MLOps Monitoring	0.18	0.020

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Assurance	0.17	0.020
Explainable AI	0.16	0.015
Human-in-the-Loop	0.15	0.015
Data Governance	0.13	0.012
Ecosystem Trust	0.11	0.010
Strategic KPIs	0.06	0.008
Data Trusts/Contracts	0.04	0.007

Table 9: BWM Robustness (Mean $\pm$ SD under Perturbation)

ISM threshold variations (± 1 level) and alternative defuzzification parameters did not alter the driver-dependent classification for any top-ranked enabler, further confirming structural stability. Bootstrap confidence intervals for key Spearman correlations (e.g., Data Governance–Assurance/Auditability) remained significant at the 95% level across all resamples, ruling out spurious associations.

Discussion

The findings provide coherent analytical support for the rationale for a stratified DI framework and elucidate how capabilities transmit value from fundamental data components to strategic and ecosystem-level outcomes. The theme analysis revealed a focus on enablers pertaining to assurance, human–AI cooperation, and governance, as well as a significant presence of Strategic KPIs and ESG-related constructs. The frequency analysis corroborates this tendency, revealing that decision-centric and governance-oriented skills are much more prevalent compared to specialized analytical approaches (Abirami & Chitra, 2021; Raghupathi & Raghupathi, 2021).

The correlation analysis reinforces this perspective by uncovering regular relationships among essential constructs. Data Governance is positively correlated with Assurance/Auditability, indicating that firms prioritising data quality and lineage are likely to include audit mechanisms, a finding aligned with the literature on responsible AI and governance (Delen & Zolbanin, 2018; Zaidan & Ibrahim, 2024). Explainable AI is correlated with both Bias/Fairness and Human-in-the-Loop, underscoring its function in fostering trust, supervision, and ethical risk management, as extensively emphasized in research on human–AI collaboration and explainability (de Oliveira et al., 2023; Schneider et al., 2023). At the ecosystem level, interoperability is linked to ecosystem trust and data trusts/contracts, indicating the co-development of technological standards and governance procedures in data-sharing contexts (Toorajipour et al., 2024; Batool et al., 2025). The identified correlation between Strategic KPIs and ESG Compliance underscores the incorporation of sustainability factors into corporate performance

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frameworks, aligning with recent studies on AI governance and sustainability integration (Papagiannidis et al., 2025).

The causal-structural analysis provides further insight into the interactions among these enablers. The DEMATEL findings reveal that MLOps/Monitoring, Assurance & Auditability, Explainable AI, Human-in-the-Loop, Data Governance, and Interoperability have a positive net effect (D–R), categorizing them as causal drivers inside the system. Conversely, Strategic KPIs, ESG Compliance, Risk Appetite, and Data Trusts/Contracts emerge as dependent variables that reinforce upstream influences. The upstream–downstream framework aligns with socio-technical viewpoints on decision intelligence, in which essential technological and governance competencies facilitate superior strategic outcomes (Simón et al., 2024; Tasleem et al., 2024). The ISM hierarchy systematically arranges these relationships into a structured progression, advancing from fundamental data capabilities through analytical and operational layers to decision-making and strategic frameworks, mirroring the phased capability development frequently noted in analytics maturity and digital transformation studies (Pappas et al., 2018; Papagiannidis et al., 2025).

The MICMAC classification enhances this understanding by differentiating among drivers, dependents, and linking factors. MLOps Monitoring, Data Governance, and Explainable AI serve as significant catalysts, while Strategic KPIs and ESG Compliance are seen as contingent results. Multiple factors, including Human-in-the-Loop, Assurance, Scenario Modeling, Ecosystem Trust, and Data Trusts/Contracts, demonstrate interrelated traits, indicating their dual function as both shaped by upstream capabilities and impactful on downstream results. This dual stance aligns with previous studies indicating that governance and human-centered procedures serve as intermediaries between technological systems and organizational results (Bao et al., 2023; Schneider et al., 2023). The disparities observed between the DEMATEL and MICMAC classifications align with the methodological literature, which underscores that both methodologies reflect complementary dimensions of system structure rather than equivalent causal functions (Godet, 2001).

The prioritizing outcomes derived from the BWM correspond with the causal findings, emphasizing MLOps/Monitoring and Assurance & Auditability as the most significant skills, followed by Explainable AI and Human-in-the-Loop. This weighted structure indicates that lifecycle management, monitoring, and assurance operations are crucial to enabling effective, reliable decision intelligence. These results align with recent research that identifies MLOps and continuous monitoring as essential for maintaining AI performance, dependability, and compliance with governance over time (Sauer & Burggräf, 2025; Wissuchek & Zschech, 2025). The reduced significance of ecological and strategic factors aligns with their designation as dependent outcomes, suggesting they depend on upstream technological and operational capabilities.

The DIRI converts these priorities into an evaluative operational instrument. The DIRI score indicates

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moderate preparedness, with somewhat lower weights for ecosystem-level processes and strategic alignment variables. This pattern aligns with the structural results, indicating that these pieces rely on the development of the foundational data, analytics, and assurance capabilities. These phased readiness evaluation methodologies align with previous research on analytics maturity and the evolution of decision-support capabilities (Schneider et al., 2023; Simón et al., 2024). To illustrate the practical applicability of the multi-layer DI framework, Table 10 presents three exemplary cross-industry case applications in manufacturing, banking, and healthcare. These cases are analytically constructed to demonstrate the operation of identified enablers and interrelationships in real-world contexts, adhering to the study's conceptual-analytical methodology. Instead of providing empirical validation, these cases systematically align organizational processes with the five framework layers, highlighting how data infrastructure, AI and analytics, human-AI interaction, strategic alignment, and governance mechanisms collectively influence decision outcomes. Across the cases, consistent patterns emerge that bolster the framework's fundamental propositions: *in manufacturing*, MLOps monitoring and data governance serve as essential catalysts for operational efficiency; *in banking*, explainable AI and auditability are pivotal in mitigating regulatory risk and ensuring equity; and *in healthcare*, human-in-the-loop and assurance mechanisms are vital for sustaining trust and accountability in high-stakes decisions. Hence, strategic and ecosystem-level outcomes, such as performance enhancement, compliance, and stakeholder trust, emerge as dependent results influenced by upstream technical and organizational facilitators, thereby offering practical support for the layered structure and illustrating how analytically derived relationships from DEMATEL and BWM can guide practical decision system design while remaining anchored in the study's literature-based foundation.

Layer	Case One Manufacturing: Predictive Maintenance	Case Two Banking: AI-Based Credit Risk Assessment	Case Three Healthcare: Clinical Decision Support System
Context & Objective	Reduce machine downtime and maintenance costs through predictive analytics	Improve loan approval accuracy while ensuring regulatory compliance and fairness	Enhance diagnostic accuracy and patient outcomes using AI-assisted recommendations

Layer 1: Data & Infrastructure	-IoT sensors: collect real-time machine data; -Cloud-based storage: ensures scalability; -Strong data governance: ensures data quality and interoperability	-Integration of customer financial, transactional, and behavioral data; -Governed under strict data privacy and regulatory frameworks (e.g., KYC, AML)	Electronic Health Records (EHR), imaging, and lab data integrated into centralized systems; strict data privacy and interoperability standards (e.g., HIPAA-like compliance)
Layer 2: AI & Analytics	-Machine learning models: predict equipment failures; -Anomaly detection algorithms: supported by XAI tools	-AI-driven credit scoring models: incorporating fairness metrics; -Bias detection and explainability tools: ensure transparency	AI models assist in diagnosis (e.g., imaging analysis); XAI tools provide interpretable insights for clinicians
Layer 3: Decision Intelligence Core	-Human engineers: validate AI-generated maintenance alerts (Human-in-the-Loop); -Simulation models test: maintenance scenarios	-Loan officers review: AI recommendations; -Decision support systems: incorporate risk thresholds and auditability mechanisms	-Physician: review AI-generated recommendations; -Human-in-the-loop: ensures clinical accountability and contextual judgment
Assurance & Auditability Mechanisms	Continuous monitoring via MLOps pipelines; audit logs track model predictions and system performance	Audit trails for decision transparency; bias monitoring and compliance reporting integrated into workflows	Model validation, bias checks, and audit logs ensure reliability and ethical compliance in clinical decisions

Layer 4: Strategic Alignment	Decisions aligned with KPIs Examples: downtime reduction, operational efficiency, and cost savings	Alignment with strategic KPIs Examples: default risk reduction, profitability, and regulatory compliance; integration of risk appetite frameworks	Alignment with patient safety metrics Examples: treatment success rates, and hospital performance indicators
Layer 5: Ecosystem & Governance	-Data-sharing agreements with equipment vendors; -Interoperability standards enable cross-system integration; -Governance ensures accountability	-Regulatory compliance (e.g., financial regulations); -Data-sharing frameworks with credit bureaus; -Governance ensures transparency and trust	-Compliance with healthcare regulations; -Collaboration across departments and external providers; -Ethical governance ensures patient trust
Key Drivers (Aligned with DEMATEL Results)	-MLOps Monitoring, -Data Governance, -Explainable AI	-Explainable AI, Assurance & Auditability, -Data Governance	-Human-in-the-Loop, -Explainable AI, -Assurance Mechanisms
Dependent Outcomes (Strategic/Ecosystem Level)	-Operational efficiency gains, -Reduced downtime, -Improved asset utilization	-Improved credit decision quality, -Reduced default rates, -Enhanced regulatory compliance	-Improved patient outcomes, -Reduced diagnostic errors, -Enhanced trust in AI-assisted care
Alignment with BWM Priorities	-Strong alignment with top-ranked enablers: MLOps Monitoring and Assurance	High alignment with Explainable AI and Assurance priorities	Strong alignment with Human-in-the-Loop and Assurance priorities
Key Insight	Demonstrates how foundational data and monitoring capabilities drive	Highlights the role of governance, fairness, and explainability in	Emphasizes the critical role of human-AI collaboration and

	operational efficiency	decision	high-stakes decisions	financial	auditability in decision environments	sensitive
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Table 10: Illustrative Cross-Industry Applications of the Multi-Layer DI Framework Demonstrating Alignment of Enablers, Analytical Drivers, and Strategic Outcomes

The results together enhance traditional analytics viewpoints by highlighting that decision intelligence maturity is not just dependent on model complexity. Rather, it signifies the integrated advancement of data governance, model lifecycle management, interpretability, human oversight, and alignment with strategic and environmental imperatives. This endorses the perspective of decision intelligence as a socio-technical system wherein technical, organizational, and governance capabilities are interrelated and organized across various layers, aligning with modern frameworks in decision intelligence and human–AI collaboration (Simón et al., 2024; Tasleem et al., 2024). Simultaneously, these interpretations must be evaluated in the context of the study's design. The analytical inputs originate from a systematic synthesis of literature rather than from raw empirical data or external expert consultation. Thus, the findings provide a clear and methodically obtained depiction of the links among enablers; yet, they do not serve as direct empirical confirmation of causal effects within organizational contexts. This constraint aligns with earlier methodological considerations about conceptual-analytical frameworks, which underscore their function in organizing issue domains and directing subsequent empirical investigations (Sauer & Burggräf, 2025). Subsequent research may expand upon this study by evaluating the framework via expert-driven analyses, industry-specific implementations, or long-term field surveys.

Conclusion

This research presents and analytically assesses a multi-layered strategic framework for AI-enhanced DI, including Data & Infrastructure → AI & Analytics → DI Core → Strategic Alignment → Ecosystem & Governance. The combined DEMATEL–ISM–MICMAC analysis reveals a coherent causal hierarchy in which fundamental data and operational competencies, such as data governance, interoperability, and MLOps—facilitate explainable, human-centric decision-making processes, ultimately leading to strategic and ecosystem-level outcomes. This stratified advancement aligns with the evolving paradigms of decision intelligence and socio-technical systems, which underscore the amalgamation of data, analytics, human discernment, and governance frameworks (Simón et al., 2024; Tasleem et al., 2024). BWM priority and the DIRI idea convert this framework into a pragmatic roadmap, with assurance-centric operations serving as a key leverage point for reliable, scalable AI implementation. The approach depends on literature-based analytical inputs instead of primary field validation, making its contribution more about conceptually rigorous design and prioritizing than actual evidence of organizational performance impacts. This research contributes to the literature in three primary ways. Initially, it provides a cohesive architecture that amalgamates technological, organizational, and ecological factors

inside a singular decision-oriented framework, enhancing previous analytics and governance models that often address these areas in isolation (Pappas et al., 2018; Papagiannidis et al., 2025). Secondly, it delineates a methodically founded sequence of capacities, elucidating the links between drivers and dependents across layers, which corresponds with and enhances causal-structural methodologies used in the study of large socio-technical systems (Godet, 2001). Third, it integrates a sustainability-focused sub-framework that embeds ESG considerations alongside conventional performance indicators, thereby enhancing the existing literature on responsible AI and governance-aligned analytics (Batool et al., 2025; Schneider et al., 2023).

This study reveals several prospects for further investigation. Future research may include the concept in prospective longitudinal field studies spanning sectors such as manufacturing, public health, and fintech to assess the empirical validity of DIRI standards in real-world contexts. Further study is required to evaluate value-of-information impacts and real-options factors in AI investment portfolios, as well as to calibrate sector-specific DIRI standards using industry-adjusted weight sets. Moreover, augmenting the ESG sub-framework with primary survey data will facilitate the assessment of the extent and configuration of the suggested trade-offs between sustainability and financial success. These initiatives would bolster the external validity of the framework and improve its application across various organizational settings, aligning with demands for empirical substantiation in decision intelligence and AI governance research (Sauer & Burggräf, 2025; Wissuchek & Zschech, 2025). The findings indicate that organizations and policymakers ought to enhance data governance and operational controls, institutionalize assurance as a fundamental capability, align AI initiatives with strategic and ESG priorities, and expand implementation via credible ecosystem governance mechanisms. These guidelines align with extensive studies that highlight that the production of sustainable AI value relies on governance, trust, and ecosystem coordination rather than solely on technological capabilities (Delen & Zolbanin, 2018; Toorajipour et al., 2024). Collectively, these measures provide a framework for transforming AI from a mere collection of technological tools into a robust, reliable decision-making system.

Funding

No funding is available for this study.

Conflict of Interest

The authors(s) have no conflict of interest to declare

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